



ECON-UA 12

Intermediate Macroeconomics

Jaroslav Borovička, NYU, Fall 2013



Topic 5

Growth and ideas

Lecture objectives and reading

- ▶ Previous lecture – Solow growth model
 - ▶ Capital accumulation alone cannot generate long-run growth.
 - ▶ Diminishing MPK implies convergence to steady state.
- ▶ Source of growth that does not suffer from diminishing MPK
 - ▶ **Ideas**: non-rivalrous goods – one person's use of an idea does not limit the availability of the idea to others.
 - ▶ Ideas are therefore conceptually different from other goods.
 - ▶ Innovation as a source of growth.
 - ▶ **Romer model** – model of technological progress.
- ▶ Tools
 - ▶ Fixed cost and increasing returns to scale $\Rightarrow$ scale effects
 - ▶ Romer model + Solow model $\Rightarrow$ long-run growth
- ▶ Reading
 - ▶ Jones – Chapter 6 + additional materials on course website
 - ▶ See also list of books on page 157 of the textbook!

Outline

1. **The economics of ideas**
 - ▶ Fixed cost of R&D, increasing returns to scale.
 - ▶ Providing monopoly power through patents.
2. **Romer model (1990)**
 - ▶ Sustained long-run growth.
3. **Romer model + Solow model**
 - ▶ Long-run growth and transition dynamics
4. **Growth accounting**

Objects versus ideas

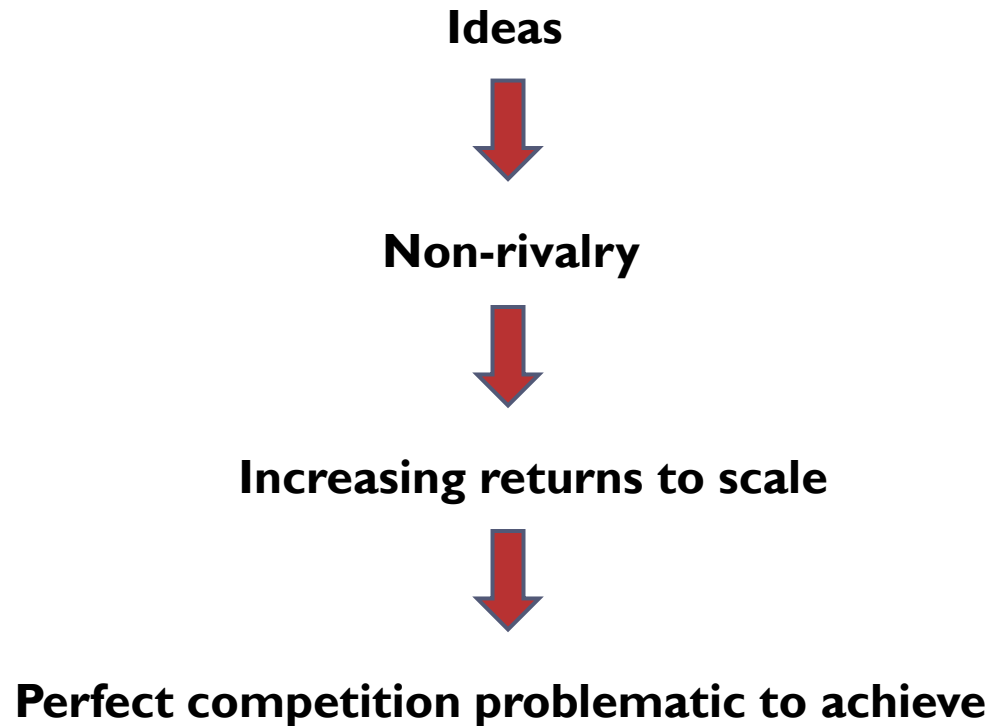
- ▶ **Objects are rivalrous**

- ▶ One person's use of an object reduces the usefulness of the object to somebody else. This holds for most goods.
- ▶ Adam Smith's **invisible hand**: strive for perfect competition, which will lead to desirable allocations.

- ▶ **Ideas are non-rivalrous**

- ▶ What are ideas: instructions, designs for making/using objects.
- ▶ Examples: computer designs, production technologies, medical treatment, management techniques, knowledge of the impact of macroeconomic policy, ...
- ▶ Many people can use the same idea independently.
- ▶ Problem: Coming up with an idea (*producing an idea*) is costly. But once the cost is paid and an idea is generated, everybody can use it $\Rightarrow$ **increasing returns to scale**.

Ideas and increasing returns to scale



Ideas and increasing returns to scale

- ▶ Imagine ideas inventors keep new ideas secret
 - ▶ Then everybody has to invent everything, and pay the cost of new ideas.
 - ▶ The same idea is discovered again and again.
 - ▶ This is too costly.
- ▶ Ideas are non-rivalrous – it is enough to discover an idea once, pay the cost once, and then share it with everybody.
 - ▶ Increasing returns to scale: once the cost is paid, the idea is available for free $\Rightarrow$ average cost per user declines with the number of users.
 - ▶ **Problem:** Who should be the one who pays the cost?
 - ▶ **Question:** Should larger economies produce more ideas?
 - ▶ More resources to produce ideas $\Rightarrow$ more ideas.
 - ▶ More users $\Rightarrow$ lower cost of idea production per user.
 - ▶ But larger countries are not richer and do not grow faster.

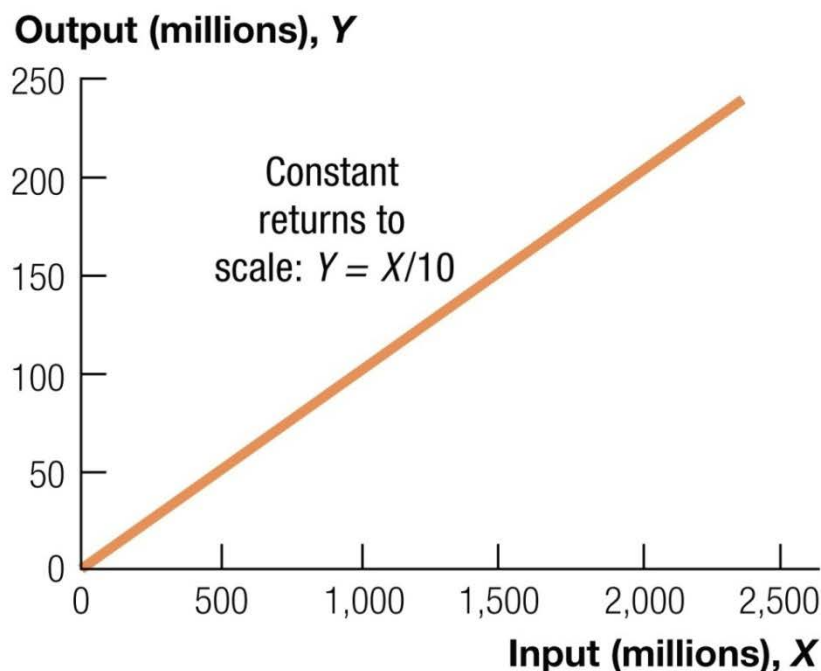
Competition and **constant** returns to scale

- ▶ With CRS, firms earn zero economic profits.
 - ▶ **Replication argument**: Doubling inputs doubles output.
 - ▶ **Rivalry**: Producing one chair does not mean everybody can now use the same chair.
 - ▶ Anybody can use the same technology.
 - ▶ If firm charges a price for its output that generates positive profits, competing firms will undercut the price to conquer its market share.
 - ▶ Undercutting stops when firms reach a price at which they earn zero economic profit.
- ▶ Same argument holds for hiring labor and capital
 - ▶ Next slide

Competition and **constant** returns to scale

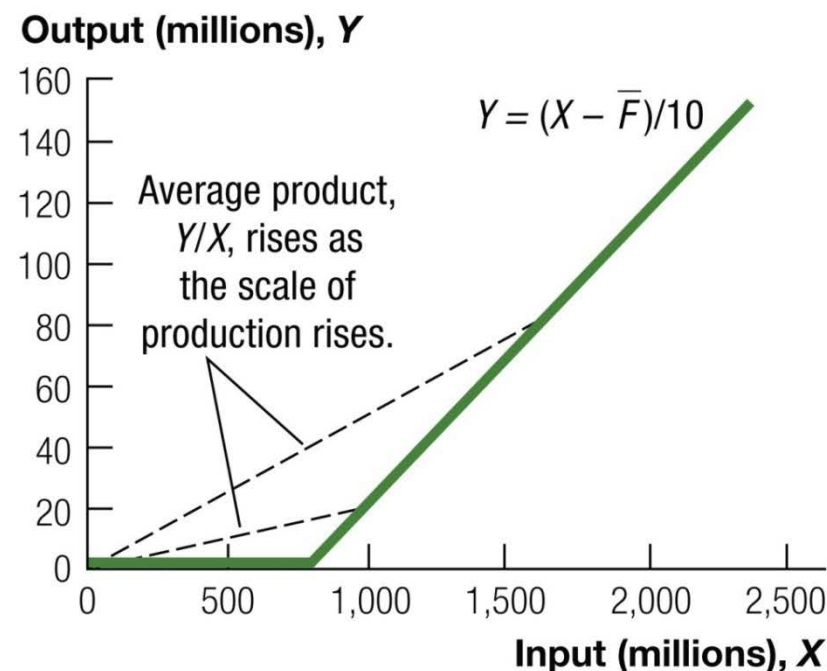
- ▶ Hiring labor (same argument holds for capital): $MPL = w$
 - ▶ Firm takes market wage w as given, chooses amount of labor to satisfy the first-order condition.
 - ▶ When wages are lower, firm **demands** more labor (**intuitive!**)
 - ▶ But how is w determined? **Competition.**
- ▶ Example 1: A **single firm** decreases wages to increase profits
 - ▶ Then workers go work somewhere else.
- ▶ Example 2: **All firms** decrease wages to increase profits
 - ▶ When wages are too low, profit-maximizing firms demand more labor than what households are willing to supply.
 - ▶ Firms start to compete for workers by offering higher wages.
 - ▶ Wages increase until labor supply = labor demand, and firms earn zero economic profits again.
- ▶ But **increasing returns to scale** are different – next slide.

Increasing returns to scale



(a) Constant returns to scale:

$$Y = X/10$$



(b) Increasing returns from fixed cost:

$$\bar{F} = 800$$

- ▶ IRS: Average cost of production decreases as scale of production increases.
- ▶ Example (b): development of a new drug
 - ▶ Costs \$800 million to develop.
 - ▶ Production costs are \$10 per dose.

Competition and **increasing** returns to scale

- ▶ Firm pays for an idea (cost of R&D)
 - ▶ **Fixed cost** paid regardless of how many times the idea is used.
 - ▶ Must earn positive profits on goods that embody the idea.
 - ▶ But if these goods can be sold at a profit, this will attract other firms that can do the same (to them, the idea is freely available).
 - ▶ Competitive firms start to undercut prices $\Rightarrow$ in equilibrium, firms will earn zero economic profits.
 - ▶ Economically, under competition, **price equals marginal cost**.
 - ▶ Marginal cost of using an idea is zero, so the firm in equilibrium will charge a **zero price for including the idea** in the product.
- ▶ But if the original inventor cannot charge anything extra for including the idea in the product, how does she recover the R&D cost?
 - ▶ She cannot! $\Rightarrow$ She will not even start with the R&D!
 - ▶ But then how do we generate inventions?

Competition, CRS, IRS – a summary

- ▶ **Goods:** CRS (replication argument) + rivalry
 - ▶ A competitive market will exist.
 - ▶ Allocation is **Pareto optimal** – you cannot make one person better off without making somebody else worse off.
 - ▶ We can redistribute from one person to another (e.g., to offset inequality) but somebody has to pay the cost.
- ▶ **Ideas:** IRS (fixed cost of R&D) + non-rivalry
 - ▶ A competitive market will not work.
 - ▶ There is not a price which would, at the same time,
 - ▶ be **competitive** (price equals marginal cost – efficient allocation of production resources), and
 - ▶ allow to **recoup R&D costs** for the inventor.
 - ▶ We need some institutions, like legal protection.

Competition and IRS – solutions?

- ▶ Grant the investor a **monopoly right** to use the idea
 - ▶ **Exclude** others from using the same idea.
 - ▶ Non-rivalry: Intrinsic/physical characteristic of an idea.
 - ▶ Excludability: A feature chosen by society (extent of property rights).
 - ▶ Implementation: patents and other intellectual property rights.
 - ▶ Investor can charge higher prices and generate profit $\Rightarrow$ covers the cost of R&D.
 - ▶ **Problem 1**: Profit-maximizing monopolies constrain output $\Rightarrow$ inefficiency.
 - ▶ Patented drugs, seeds, ...
 - ▶ **Problem 2**: Most of research is incremental – new ideas build on old ideas. The sole owner of an idea can inhibit future innovations.
 - ▶ Semiconductors, Apple vs. Samsung, ...

Competition and IRS – solutions?

- ▶ What is the right balance?
 - ▶ The promise to grant a patent should promote innovation activity **BEFORE** the idea is discovered.
 - ▶ But **AFTER** the idea is discovered, we want to limit monopoly power.
 - ▶ **Tradeoff**: Does the current degree of patent protection balance out the benefits of innovation and cost of monopoly power appropriately?
- ▶ This is a very difficult question:
 - ▶ How to measure the benefits and costs of innovation?
 - ▶ Case studies: Does increasing the patent protection increase innovation activity?
 - ▶ Very weak evidence that this indeed happens.

Intellectual property rights – criticism

1. Many changes in intellectual property rights target the **strengthening of protection of existing ideas**.
 - ▶ Example: Copyright Term Extension Act (“Disney Law”)
 - ▶ This makes no sense: Protecting ideas that have been already discovered only strengthens monopoly power.
2. There are large **administrative costs**.
 - ▶ Lawsuits, law-making, lobbying, patent applications, research to avoid patent violations ...
3. Inventor can benefit from a new ideas **even without patent protection**
 - ▶ Competitors need time to learn about the idea, get to understand it and implement it $\Rightarrow$ enough time for the inventor to recoup the R&D cost.
 - ▶ Most inventions and ideas are never patented, and they still happen.

Example: Patenting “obvious” ideas

- ▶ From the newspaper (PCWorld, Feb 12, 2013):
 - ▶ “Facebook sued over 'Like' button by Dutch programmer's family”
 - ▶ “The lawsuit alleges Facebook's "Like" button and other content-sharing features infringe on the patents. The patents in question were granted in 2001 and 2002 to Joannes Jozef Everardus Van Der Meer, a computer scientist. ... He died, however, in June 2004. Since then, his widow and family have pursued compensation for his inventions ...”
- ▶ Questions
 - ▶ Is a ‘Like’ button an invention?
 - ▶ Did developing the original idea bear large fixed costs?
 - ▶ Is the idea trivial/obvious? Is the idea of a “Like” button alone useful?
 - ▶ Who (if anybody at all) should be compensated?
 - ▶ Nobody disputes private property of goods. Are ideas the same? Should a patent therefore have an infinite length?

Example: Shameless copying or beneficial and necessary technological convergence?

Solow growth model



375,000 RESULTS Any time ▾

[Neoclassical growth model](#) - Wikipedia, the free encyclopedia

[en.wikipedia.org/wiki/Solow_growth_model](#) ▾

Development of the model · Graphical ... · Mathematical framework

The neoclassical **growth model**, also known as the **Solow–Swan growth model** or exogenous **growth model**, is a class of economic **models** of long-run economic **growth** ...

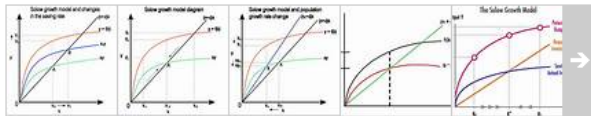
[SOLOW GROWTH MODEL](#) - Department of Economics

[www.econ.iastate.edu/.../Spring2006/SOLOW/SOLOWGROWTHMODEL.htm](#) ▾

SOLOW GROWTH MODEL. Start with a Constant Returns to Scale (CRTS) production function: $Y = f(K, L)$. CRTS implies that by multiplying each input by some factor "z" ...

[Images of Solow growth model](#)

[bing.com/images](#)



[The Solow Growth Model](#) - University of Pittsburgh

[www.pitt.edu/~mgahagan/Solow.htm](#) ▾

The **Solow** per capita production function The production function **model** was applied to the study of **growth** problems by Robert **Solow** (American economist, ...

[exegesis](#)

[patunru.blogspot.com](#) ▾

But to my surprise, the key reference of all these works (and hence mine), i.e. **Solow's** neoclassical **growth model** is not on the list. Posted by Aco at

Related searches for **Solow growth model**

[Physical Capital per Worker Formula](#)

[Solow Growth Model Golden Rule](#)

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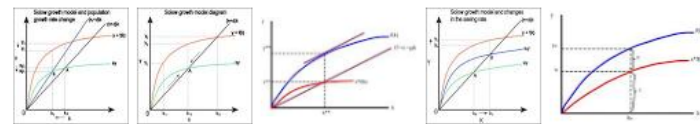
[Neoclassical growth model](#) - Wikipedia, the free encyclopedia

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Development of the model · Graphical representation of ... · Mathematical framework

[Images for solow growth model](#) - Report images



[Solow growth model](#)

[www.pitt.edu/~mgahagan/Solow.htm](#)

The production function **model** was applied to the study of **growth** problems by Robert **Solow** (American economist, Massachusetts Institute of Technology, Nobel ...

[SOLOW GROWTH MODEL](#)

[www.econ.iastate.edu/.../SOLOW/SOLOWGROWTHMODEL.htm](#)

SOLOW GROWTH MODEL. Start with a Constant Returns to Scale (CRTS) production function: $Y = f(K, L)$. CRTS implies that by multiplying each input by some ...

[\[PDF\] The Solow Growth Model](#)

[www.unc.edu/~jbhill/Solow-Growth-Model.pdf](#)

File Format: PDF/Adobe Acrobat - [Quick View](#)

The **Solow Growth Model** is a model of capital accumulation in a pure production economy: there are no prices because we are strictly interested in output = real ...

One is Google, one is Microsoft... Which one is which?

Intellectual property rights – criticism

- ▶ Some statements to think about
 - ▶ The number of patents and patent applications grows very fast.
 - ▶ Patents expand into new areas (software, ...) where **novelty** and **non-obviousness** are difficult to prove.
 - ▶ Many firms spend a lot of resources on building patent portfolios to gain bargaining power against other patent holders.
 - ▶ Is this a socially beneficial activity?
- ▶ Boldrin and Levine – *Against Intellectual Monopoly*
 - ▶ Optimal patent length may not be zero, but given the large administrative cost, it would be better to get rid of patents altogether, except industries with extremely high R&D costs and legal requirements (healthcare).
 - ▶ Inventors often collect enough profits by innovating very fast, so that they are always ahead of the imitators $\Rightarrow$ no need for additional legal protection.
 - ▶ Pretty radical idea. They provide a lot of economic arguments and empirical evidence but the debate is far from over.

Intellectual property rights – alternatives

- ▶ Competing for innovation
 - ▶ **Prizes** for best innovation – DARPA challenge, X-Prize
 - ▶ Maybe suitable for radically new innovations (space flight).
- ▶ Government **buys out developed ideas** and provides them to the public
 - ▶ How to prize newly developed ideas appropriately? Very difficult, especially for the government who is not an expert.
 - ▶ **Example:** Orphan drugs (currently, the government does not buy drugs out but provides special incentives to develop treatments for rare disorders)
- ▶ Direct research subsidies (NSF, ...)
- ▶ See case studies in the textbook

Diverse incentives to conduct research

- ▶ Consider some revolutionary discoveries
 - ▶ Penicillin and some other important drugs
 - ▶ Steam engine, combustion engine, electricity
 - ▶ Internet
- ▶ Compare the overall **social return** with the **private returns** to the inventor.
 - ▶ Are these comparable? Was the inventor rewarded sufficiently?
- ▶ **Open source software** – producing free software has many benefits to the authors other than a direct monetary reward
 - ▶ Many alternative ways for inventors to be rewarded.
 - ▶ Maybe more difficult with very costly R&D
 - ▶ but observe that **globalization** actually helps here – larger market means it is possible to recoup R&D costs more easily and faster.
- ▶ Innovate or imitate? Sometimes it is hard to tell apart.

IPR and developing countries

- ▶ Countries that produce new ideas have incentives to protect them worldwide.
- ▶ But many developing countries achieve high growth also partly due to infringement of intellectual property rights.
- ▶ **Example 1:** United States in 18th-19th century
 - ▶ Famous for illegally reprinting works of English authors.
- ▶ **Example 2:** India today
 - ▶ Patent applications rise quickly but only a small fraction comes from domestic companies, most from multinationals.
 - ▶ Whom do the patents protect? Who extracts rents?
 - ▶ See “**Imitate or Die**” article on reading list.
- ▶ Large economic literature on **catch-up growth through imitation**. Poor countries can benefit from adoption of ideas produced in rich countries.
- ▶ If we are after a global system of IPR protection, shouldn't we take into account global welfare?

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4. Growth accounting

Infinite growth on a finite Earth

- ▶ Ever heard about criticism of “infinite growth” theories?
 - ▶ The criticism concerns economic models that use a simplifying assumption of an infinite time horizon and constant growth rates.
 - ▶ Argument: Earth is finite, growth therefore cannot be infinite.
- ▶ Largely indicates a lack of understanding of economics
 1. **Economic modeling argument:** the infinite horizon is a simplification – maybe growth will stop at some distant point in the future but it is so distant that it is irrelevant today.
 2. **Empirical argument:** Growth does not mean just more steel, cars, and bigger houses. Developed economies today grow mainly through technological progress.
 - ▶ Better cars, smaller and faster computers, ...
 - ▶ Physical resources may be finite, but **ideas are virtually unlimited.**

Romer model (1990)

- ▶ Distinction between ideas and objects
- ▶ Output is produced by combining ideas and labor
- ▶ Ideas are non-rivalrous
 - ▶ No diminishing marginal returns.
 - ▶ Long-run growth with finite resources through accumulation of ideas possible – new ideas allow to use the finite resources in better and better ways.
 - ▶ **Example:** Sand – from being useless to glass to semiconductors
- ▶ Population divided into two groups
 - ▶ Producers of ideas (“researchers”)
 - ▶ Producers of output (“workers”)

Framework of the model

- ▶ Discrete time: $t = 0, 1, 2, 3, \dots$
- ▶ Initial stock of ideas (initial condition): A_0
- ▶ Resource constraint: Fixed labor force $\bar{L}$ divided between L_{yt} workers and L_{at} researchers: $L_{yt} + L_{at} = \bar{L}$
- ▶ Workers produce output: $Y_t = A_t L_{yt}$
 - ▶ A_t is the stock of existing ideas (more ideas in place make workers more productive)
- ▶ Researchers produce new ideas: $\Delta A_{t+1} = \bar{z} A_t L_{at}$
 - ▶ $\bar{z}$ is the productivity of the idea-producing technology
 - ▶ No diminishing marginal product of ideas (persistent growth!)
 - ▶ Increasing returns in the stock of ideas and number of researchers
- ▶ A constant fraction of labor $\bar{l}$ is allocated to production of new ideas: $L_{at} = \bar{l} \bar{L}$

Framework of the model

► Variables

- Stock of ideas A_t
- # of workers L_{yt}
- # of researchers L_{at}
- Output Y_t

4 equations
4 variables



► Parameters

- Share of labor force in research $\bar{l}$
- Size of labor force $\bar{L}$
- Productivity $\bar{z}$
- Initial amount of ideas A_0

► Equations

- Output production:

$$Y_t = A_t L_{yt}$$

- Accumulation of ideas:

$$\Delta A_{t+1} = \bar{z} A_t L_{at}$$

- Resource constraint:

$$L_{yt} + L_{at} = \bar{L}$$

- Labor allocation:

$$L_{at} = \bar{l} \bar{L}$$

Model solution

- ▶ Crucial component is the production technology

$$y_t = \frac{Y_t}{\bar{L}} = A_t(1 - \bar{l})$$

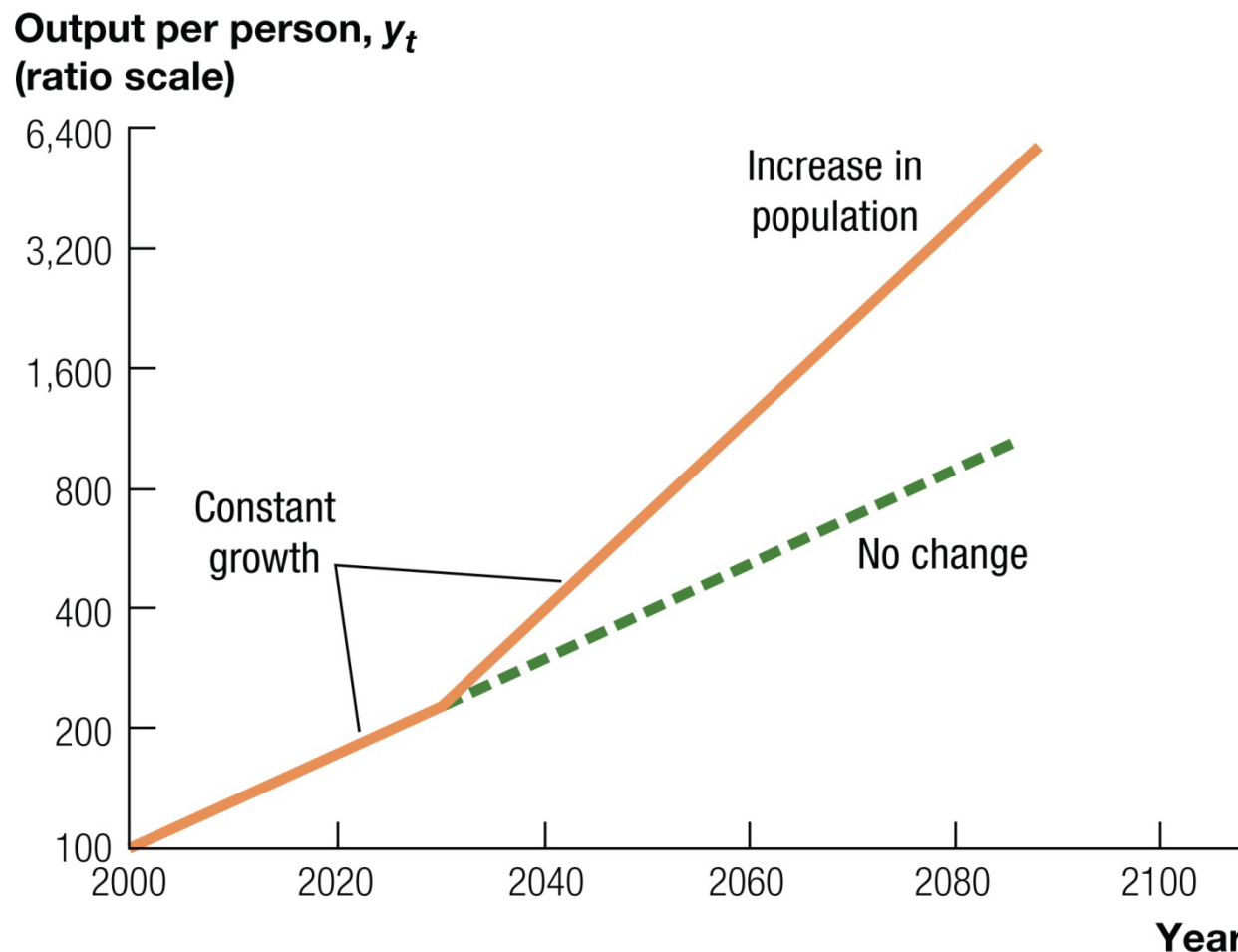
- ▶ Output per capita depends on the stock of ideas, not on the stock of ideas per capita (compare Solow $y = Ak^{1/3}$)
- ▶ Accumulation of ideas does not slow down

$$\frac{\Delta A_{t+1}}{A_t} = g_A = \bar{z}\bar{l}\bar{L}$$

- ▶ Growth of output the same as growth of ideas: $g_Y = g_A$.
- ▶ A larger population and a larger share of population devoted to research implies faster growth.
- ▶ **Balanced growth path**: ideas and output grow at a constant rate.

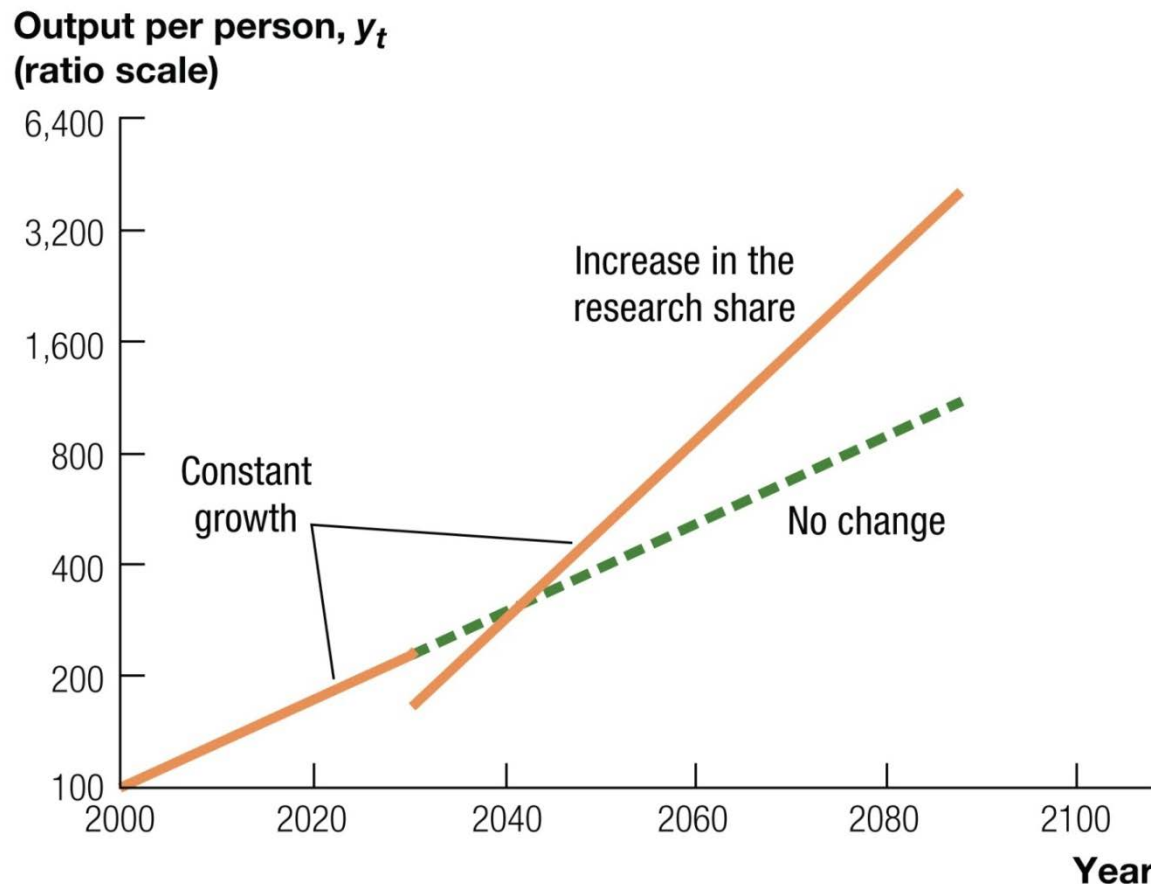
Experiments – increase in population

- Increase in population increases the growth rate



Experiments – increase in share of researchers

- ▶ Initial reallocation to research decreases the **level** of output
- ▶ Higher **growth rate** from that point on.



Romer model – discussion

- ▶ Should larger countries grow faster?
 - ▶ The model says yes.
 - ▶ But this is not empirically correct.
 - ▶ Maybe ideas are shared globally, not within countries $\Rightarrow$ model of global knowledge.
- ▶ What if $Y_t = A_t^\beta L_{yt}$ with $\beta < 1$?
 - ▶ Still sustained growth. What is the growth rate of output now?
 - ▶ The source of persistent growth lies in accumulation of ideas
 - ▶ $\Delta A_{t+1} = \bar{z} A_t L_{at}$ no diminishing marginal returns of ideas
 - ▶ Increasing returns to scale in ideas and researchers $\Rightarrow$ more researchers can accumulate ideas at a faster rate.
- ▶ What is the role of globalization in the Romer model?
 - ▶ Sharing of ideas, gains from trade, more technologies generated in developing economies.

Solow vs Romer model

▶ Solow model

- ▶ Capital is rivalrous.
- ▶ Therefore capital per person matters for growth.
- ▶ Capital per person is subject to diminishing returns.
- ▶ Constant returns to scale – size of population does not matter for growth.

▶ Romer model

- ▶ Ideas are non-rivalrous.
- ▶ Total stock of ideas matters for growth (anybody can use any idea).
- ▶ Accumulation of ideas not subject to diminishing returns $\Rightarrow$ sustained growth.
- ▶ Increasing returns to scale – larger economies more productive.
- ▶ No transition dynamics – always on a balanced growth path.

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Combining Solow and Romer models

- ▶ Advantages of the two models:
- ▶ Solow model
 - ▶ Capital accumulation – transition dynamics
- ▶ Romer model
 - ▶ Accumulation of ideas – sustained growth
- ▶ Combine these two models
 - ▶ Accumulation of ideas will determine the **long-run trend** (growth rate of most developed countries).
 - ▶ Capital accumulation will determine transition dynamics around this long-run trend (catching-up periods).
 - ▶ Each country will have a steady state **relative** to the growing frontier (determined by factors like institutions).

Framework of the combined model

- ▶ Discrete time: $t = 0, 1, 2, 3, \dots$
- ▶ Initial stock of ideas A_0 , initial stock of capital K_0
- ▶ Resource constraint: Fixed labor force $\bar{L}$ divided between L_{yt} workers and L_{at} researchers: $L_{yt} + L_{at} = \bar{L}$
- ▶ Workers produce output: $Y_t = A_t K_t^\alpha L_{yt}^{1-\alpha}$
- ▶ Researchers produce new ideas: $\Delta A_{t+1} = \bar{z} A_t L_{at}$
- ▶ Capital accumulation: $\Delta K_{t+1} = s Y_t - \delta K_t$
- ▶ A constant fraction of labor $\bar{l}$ is allocated to production of new ideas: $L_{at} = \bar{l} \bar{L}$

Framework of the model

► Variables

- Stock of ideas A_t
- Stock of capital K_t
- # of workers L_{yt}
- # of researchers L_{at}
- Output Y_t

5 equations

5 variables



► Parameters

- Share of researchers $\bar{l}$
- Size of labor force $\bar{L}$
- Productivity $\bar{z}$
- Initial amount of ideas A_0
- Initial stock of capital K_0

► Equations

- Output production:

$$Y_t = A_t K_t^\alpha L_{yt}^{1-\alpha}$$

- Accumulation of ideas:

$$\Delta A_{t+1} = \bar{z} A_t L_{at}$$

- Accumulation of capital:

$$\Delta K_{t+1} = s Y_t - \delta K_t$$

- Resource constraint:

$$L_{yt} + L_{at} = \bar{L}$$

- Labor allocation:

$$L_{at} = \bar{l} \bar{L}$$

Model solution

- ▶ Determine the long-run growth rate

- ▶ Pinned down by the growth rate of the stock of ideas

$$\frac{\Delta A_{t+1}}{A_t} = g_A = \bar{z}\bar{l}\bar{L}$$

- ▶ Determine the growth rate in transition dynamics

- ▶ Define output and capital scaled by (a suitable power of) stock of ideas

$$\tilde{Y}_t = \frac{Y_t}{(A_t)^{\frac{1}{1-\alpha}}} \quad \tilde{K}_t = \frac{K_t}{(A_t)^{\frac{1}{1-\alpha}}}$$

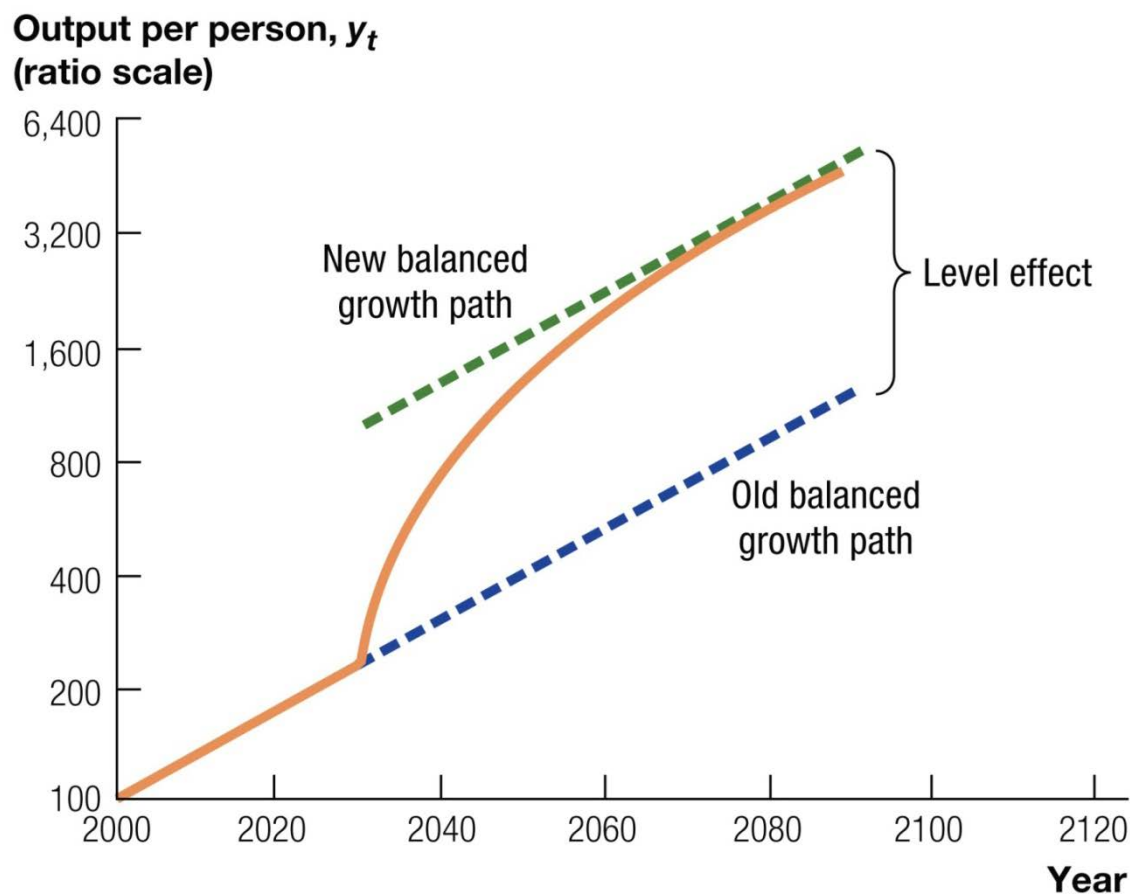
- ▶ Rewrite output and capital accumulation equations in terms of $\tilde{Y}_t, \tilde{K}_t$
 - ▶ This yields the same accumulation dynamics as in Solow model.
 - ▶ We can analyze transition dynamics using $\tilde{Y}_t, \tilde{K}_t$ instead of Y_t, K_t and then rescale back by the growing stock of ideas.
 - ▶ See derivation in class.

Model solution II.

- ▶ Plot the Solow diagram in variables $\tilde{Y}_t, \tilde{K}_t$
 - ▶ Plot the transition path for the adjustment in capital
 - ▶ Steady state: $\tilde{K}_t$ constant $\Rightarrow K_t$ and $(A_t)^{\frac{1}{1-\alpha}}$ grow at the same rates = **balanced growth path**
 - ▶ $\tilde{K}_t$ below the steady state = the economy has little capital relative to the stock of ideas
 - ▶ Positive growth rate of $\tilde{K}_t$ = capital grows faster relative to the stock of ideas, compared to the balanced growth path
 - ▶ Plot the transition path for $\tilde{Y}_t, \tilde{K}_t$, then rescale back by $(A_t)^{\frac{1}{1-\alpha}}$
 - ▶ See graphs in class and in the textbook.

Experiment – increase in saving rate

- Decompose this problem into two – **long-run growth trend** and **transition dynamics**



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Growth accounting – balanced growth path

- ▶ Growth rate of output $Y_t = A_t K_t^\alpha L_{yt}^{1-\alpha}$
 $g_{Yt} = g_{At} + \alpha g_{Kt} + (1 - \alpha) g_{Lyt}$
- ▶ Growth rate of ideas: $g_{At} = g_A = \bar{z}\bar{l}\bar{L}$
- ▶ Growth rate of capital: $g_{Kt} = s \frac{Y_t}{K_t} - \delta$
 - ▶ $\Rightarrow$ Growth rate of capital must be the same as growth rate of output.
 - ▶ Why?
- ▶ Growth rate of population – let us assume it is zero.
- ▶ Therefore: $g_{Yt} = \bar{z}\bar{l}\bar{L} + \alpha g_{Yt}$ which implies $g_{Yt} = \frac{1}{1-\alpha} \bar{z}\bar{l}\bar{L}$
- ▶ **Observation:** Output grows faster than stock of ideas.
 - ▶ Why? Capital accumulation helps out.

Growth accounting – United States

	1948–2008	1948–73	1973–95	1995–2008
Output per hour, Y/L	2.5	3.3	1.5	2.6
Contribution of K/L	0.9	0.9	0.7	1.0
Contribution of labor composition	0.2	0.2	0.3	0.2
Contribution of TFP, A	1.4	2.2	0.5	1.4

- ▶ See derivation in class / textbook
- ▶ Per hour output grows at about 2.5% per year
- ▶ Productivity slowdown during 1970s
- ▶ “New economy” since the 1990s
- ▶ What about future years?
 - ▶ Book: Cowen – The Great Stagnation

A word of caution – markets / governments

- ▶ We saw that the market need not choose the optimal level of R&D (IRS + non-rivalry)
- ▶ Can government help?
 - ▶ In principle yes, but it may be difficult in practice.
 - ▶ Optimal patent length weighs benefits of innovation with cost of monopoly.
 - ▶ Patent protection may involve very high administrative cost.
- ▶ What about direct subsidies to research?
 - ▶ Direct monetary rewards are not the only incentive.
 - ▶ Many discoveries are often unintentional.
- ▶ We still understand that innovation is crucial.
 - ▶ But it should be also closely related to human capital accumulation (left out from our analysis).
- ▶ Free-riding – how to distinguish imitation from innovation?

Review questions

- ▶ What is the distinction between objects and ideas?
- ▶ Why is the distinction important from the point of modeling of economic growth?
- ▶ Why do increasing returns and perfect competition interfere?
- ▶ What is catch-up growth through imitation?
- ▶ Are there too many or too few resources devoted to discovering new ideas?
- ▶ How does population size influence the rate of growth in the Romer model? Do we see larger countries grow faster? Why or why not?